APPLICATION NOTE

AN1422: Si5334/Si5335/Si5338 Frequently Asked Questions

This document answers some of the most frequently asked Si5334/Si5335/Si5338 questions in one place. If you do not find the answers you are looking for, please contact Skyworks Support and Resources.

Product Family

Si5334/Si5335/Si5338: General Purpose Clock Generator products

Table 1. Frequently Asked Question Links

Cross-Reference Link
1. Where can I find Si5334-, Si5335-, and Si5338-related collateral?
2. What are the main differences between the Si5334, Si5335, and Si5338?
3. Where is the RoHS and REACH compliance information for Si5334/Si5335/Si5338?
4. How can I obtain a Si5334, Si5335, or Si5338 with a custom startup frequency?
5. What is the default I2C address of the Si5338 when the part does not have an I2C_LSB pin or a custom address?
6. How can I burn the NVM on a blank Si5338 part?
7. Does the ClockBuilder Pro Field Programmer (CBPROG-DONGLE) work with the Si5334, Si5335, or Si5338?
8. Where can I find software for the Si5338 Field Programmer?
9. How can I use a ClockBuilder Pro project file with the Field Programmer?
10. Does the Si5338 support in-circuit NVM programming?
11. Can I import a Si5338 ClockBuilder desktop NVM file into ClockBuilder Pro?
12. How do I control the Si5338-EVB?
13. What if I want a feature in my custom Si5334, Si5335, or Si5338 that is not configurable in ClockBuilder Pro?
14. What power supply sequence/delay is required for VDD and VDDOx?
15. What should unused VDDOx pins be connected to?
16. When an LOS or LOL occurs, are the outputs squelched?
17. What do I need to communicate with the Si5338 using I2C?
18. How do I configure Si5338 using I2C?
19. What is the I/O delay of the Si5335 in ZDB and NZDB modes?
20. Can the Si5335 and Si5338 clock generators support spread spectrum inputs?

1. Where can I find Si5334-, Si5335-, and Si5338-related collateral?

Table 2. Product Links

Link Type	Link
Product pages	https://www.skyworksinc.com/Products/Timing/General-Purpose-Clock-Generators
Evaluation board (EVB)	https://www.skyworksinc.com/Products/Timing/Evaluation-Kits/clock/si5338-evaluation-kit
Reference manual (Si5338)	Si5338 Reference Manual
Field programmer (Si5338 only)	https://www.skyworksinc.com/Products/Timing/Evaluation-Kits/clock/si5338-56-prog-evb-evaluation-kit
ClockBuilder® Pro software	https://www.skyworksinc.com/en/Application-Pages/Clockbuilder-Pro-Software
IBIS models	Available on the specific product pages
Schematic footprint and symbols	Available on the specific product pages
Request technical support	Skyworks Support and Resources

Table 3. Relevant Application Notes

Link Type	Link
Application Note	AN360: Crystal Selection Guide for Si5334/35/38 and Si5355/56 Devices
	AN408: Termination Options for Any-Frequency, Any-Output Clock Generators and Clock Buffers
	AN428: JumpStart: In-System, Flash-Based Programming for Skyworks Solutions Timing Products
	AN473: Configuring the Si5338 Evaluation Board in Zero Delay Mode
	AN624: Si5335 Solves Timing Challenges in PCI Express, Computing, Communications and FPGA-Based Systems

2. What are the main differences between the Si5334, Si5335, and Si5338?

The **Si5334** is a pin control-only part, with pre-defined pinouts depending on the selected part number (Si5334A, Si5334B, etc.). See page 23 of the Si5334 data sheet for more details on the different available part numbers.

The **Si5335** is also a pin control-only part, however it has some additional functionality compared to the Si5334 in that it can select either 1.6 MHz or 475 kHz as the PLL loop bandwidth. See page 35 in the Si5335 data sheet for an in-depth description of the two options. AN624 has additional, application-specific details on how to select the proper bandwidth.

The Si5335 offers up to three pin-controlled custom frequency profiles that can be set up using ClockBuilder Pro. The finished part uses up to two frequency select pins to select between the different configurations.

The **Si5338** is a fully I^2C-programmable clock generator. See the Si5338 data sheet and the Si5338 reference manual for detailed programming instructions.

3. Where is the RoHS and REACH compliance information for Si5334/Si5335/Si5338?

Please refer to the part number search on the website for the Certificate of Conformance.
https://www.skyworksinc.com/product_certificate.aspx

4. How can I obtain a Si5334, Si5335, or Si5338 with a custom startup frequency?

You can use ClockBuilder Pro to create a custom part number for all three products.

After creating a custom plan, you have the option to "create a custom part number" under the **Skyworks Cloud Services** tab. This officially creates the part number in our system. After this step, the custom part number is orderable from your local distributor. Please contact your local Skyworks sales representative if you have questions.

In addition to the ClockBuilder Pro customization, the Si5334 has pre-defined "standard frequency plan" part numbers that can be found near the end of the data sheet. These are all orderable from your local distributor.

5. What is the default I²C address of the Si5338 when the part does not have an I2C_LSB pin or a custom address?

The default address of the Si5338 is always 0x70h.

The I2C_LSB pin is an option for the Si5338A/B/C/K/L/M part numbers only. All other Si5338 devices either have 0x70h as their default address, or they can have a custom I²C address defined by a custom configuration from ClockBuilder Pro.

When creating a custom part number in ClockBuilder Pro for the Si5338, you have the option to define a new I²C address.

6. How can I burn the NVM on a blank Si5338 part?

Burn the Non-Volatile Memory (NVM) of a blank, loose Si5338 using the Si5338/Si5356 field programming evaluation kit.

Once a blank Si5338 has been programmed, it can then be removed from the field programmer and mounted on a PCB for prototyping purposes.

Note: *Pre-programmed Si5338 devices cannot be re-programmed using the field programmer. It is only designed to be used with loose, blank parts for non-production sample purposes only.*

After a profile has been vetted and is ready for production, use ClockBuilder Pro to create a new custom part number for that profile. After a part number is created, you can contact your local distributor to place an order for that new part number containing the new custom profile.

7. Does the ClockBuilder Pro Field Programmer (CBPROG-DONGLE) work with the Si5334, Si5335, or Si5338?

The CBPROG-DONGLE supports volatile I²C communication with a Si5338 in-system. Connect the GND, SCLK, and SDA_SDIO pins from the dongle to the GND, SCL, and SDA pins on the Si5338. ClockBuilder Pro can then be used to modify individual volatile registers or write full project files to the Si5338.

Note: *The CBPROG-DONGLE cannot be used to burn the NVM of an in-system or loose Si5334, Si5335, or Si5338. It only supports NVM programming of the Si538x, Si534x/9x, Si5332, etc.*

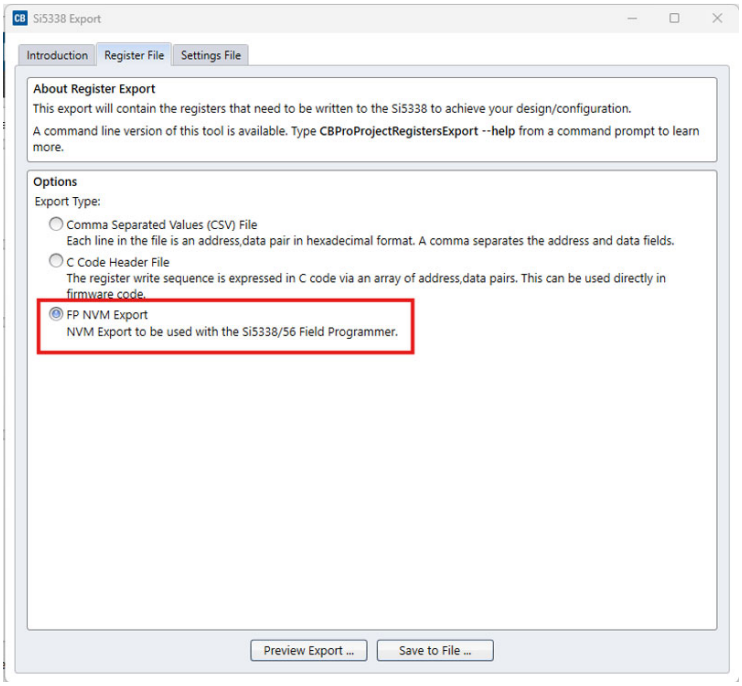
Use the Si5338/Si5356 Field Programmer (Si5338-PROG-EVB) to burn the NVM of loose Si5338 samples.

8. Where can I find software for the Si5338 Field Programmer?

ClockBuilder Pro now supports burning loose parts with the Si5338/56 Field Programmer.

9. How can I use a ClockBuilder Pro project file with the Field Programmer?

1. In the Si5338 ClockBuilder Pro Project file, go to **Export** > **Register File**, then click **FP NVM Export** to create and save your NVM file.



2. With the Si5338/Si5356 Field Programmer plugged in, open the field programmer GUI through the ClockBuilder Pro design dashboard.
3. Open the recently saved NVM file in the GUI and click **Write NVM**. Your part is burned to match your ClockBuilder Pro project file.

10. Does the Si5338 support in-circuit NVM programming?

No, the Si5338 NVM cannot be in-circuit programmed. It only supports volatile I^2C programming, which is reset to the factory defaults when power is cycled.

The Si5338/56 Field Programmer is only designed to be used with loose, blank Si5338 parts. It is not designed to burn a pre-soldered Si5338.

One option for in-system I^2C programming is to use a flashed microcontroller to automatically configure Si5338 on power up. See AN428 for a full description of this solution.

11. Can I import a Si5338 ClockBuilder desktop NVM file into ClockBuilder Pro?

Yes. Follow these steps:

1. Click the “Convert Existing Project/NVM File” link in the main Clock Builder Pro Wizard page.
2. At the top of the next window, select the “Si5338 CBDesktop NVM File” and click **Next**.
3. Find and import the ClockBuilder Desktop NVM file in question.

The full original ClockBuilder desktop configuration is now available in the ClockBuilder Pro design dashboard. You can now edit the configuration and create a new custom part number.

12. How do I control the Si5338-EVB?

ClockBuilder Pro is the recommended software to use alongside the Si5338-EVB. When an EVB is plugged into a computer using the USB cable, ClockBuilder Pro recognizes the EVB and allows you to write any existing Si5338 project file to the installed Si5338 device.

Additional information on how to use the Si5338-EVB can be found in the Si5330/34/35/38 Evaluation Board User's Guide.

13. What if I want a feature in my custom Si5334, Si5335, or Si5338 that is not configurable in ClockBuilder Pro?

If ClockBuilder Pro is currently missing a configuration option that you would like included in your custom part, create a new support ticket and we can add your new settings for you.

When contacting support, remember to include your existing ClockBuilder Pro Project and a description of the functionality you would like added. We can then send a new updated ClockBuilder Pro Project file back to you. Submit the new modified configuration for a new part number that includes your required functionality.

14. What power supply sequence/delay is required for VDD and VDDOx?

None, unless a pre-programmed phase offset is expected between outputs. In that case, the VDDOx output supply voltage must cross 1.2 V before the VDD (pins 7, 24) core power supply voltage crosses 1.45 V.

15. What should unused VDDOx pins be connected to?

Tie all unused VDDOx pins to VDD.

16. When an LOS or LOL occurs, are the outputs squelched?

No, the output clocks continue to toggle even with a LOS or LOL interrupt. Without a reference, the frequencies can drift up to 5 percent in clock generator mode.

One way around this is to tie the interrupt pin to an OEB pin. When the pin goes high due to an interrupt, the OEB pin is also pulled high, disabling the outputs. Since the interrupt pin is open drain, it is important to include the suggested 1 to 4 kΩ pullup.

17. What do I need to communicate with the Si5338 using I²C?

You can use any I²C bus master that allows both read and write cycles over the I²C bus (e.g., MCU or FPGA). The basic Si5338 I²C configuration process requires reading registers (sometimes referred to as RAM in the data sheet), modifying the data programmatically, and then writing a new value. As long as the I²C bus master adapter can perform these operations, you can configure the device. I²C can be configured on any Si5338 device variant with I²C bus support, whether it is a custom part or blank part.

Creating a configuration to load into registers (RAM) is different than creating a NVM configuration. Please refer to the Si5338 data sheet, sections 3.5.2 and 3.5.3. For hardware design related data specific of our I²C interface, refer to Si5338 data sheet Table 15, "I²C specifications (SCL, SDA)."

Details about the I²C configuration procss and sample C code can be found in AN428. Skyworks does not provide any software drivers or HDL code for FPGAs.

18. How do I configure Si5338 using I²C?

There are three main documents to use as a reference in understanding the I²C configuration process. If you are new to the I²C configuration process, please review these closely. You must follow the complete process described in the documents listed below for the device to work properly.

1) Si5338 data sheet:

Read sections 3.5.1, 3.5.2, and 3.5.3, and see Figure 9. Section 3.5.4 describes field programmer for NVM prototype parts programming. These sections of the data sheet outline the process that must be followed when loading a new configuration.

Note: *The entire register map must be written when making configuration changes. Do not change individual registers based on looking at file differences when changing or modifying configurations. This can lead to improper configuration of the device.*

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2) AN428: Jump Start for In-System, Flash-Based Programming:

AN428 describes in-system I²C configuration using a microcontroller (MCU). It contains C source code for our C8051F3xx MCU family. The code specific to performing the Si5338 configuration starts on page 23 of this document and it implements the flow chart shown in Figure 9 of the Si5338 data sheet (this code can be used as a guide to implement your own code). The data array required for this code, which contains the register settings plus the write-allowed mask, can be automatically generated using ClockBuilder Pro (see the Si5338 data sheet, section 3.5.2, item 1.g.).

Importantly, please closely adhere to, and do not deviate from, the event sequences indicated in the Figure 9 flowchart as well as the required time delays. You must use the "write-allowed" masks when writing to registers because some registers contain bits that should not be modified and can't be determined beforehand. Performing the required read/modify/write register operation is essential.

3) Si5338 reference manual:

This document explains how the device works with detailed register descriptions. It contains a "from scratch" description of how to configure the device without using ClockBuilder Pro. However, we strongly recommend leveraging ClockBuilder Pro and the C code in AN428 if possible.

19. What is the I/O delay of the Si5335 in ZDB and NZDB modes?

T_{PROP} refers to the I/O delay in non-zero-delay buffer mode (also known as PLL bypass). This spec does not apply to zero delay buffer mode since the PLL is in use. The following table from the Si5335 data sheet provides a T_{PROP} value.

Table 4. Performance Characteristics

(V_{DD} = 1.8 V –5% to +10%, 2.5 V ±10%, or 3.3 V ±10%, T_A = –40 to 85 °C)

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
PLL acquisition time	t_{ACQ}	1.6 MHz loop bandwidth	—	—	25	ms
PLL tracking range	f_{TRACK}	475 kHz or 1.6 MHz loop bandwidth	5000	20000	—	ppm
PLL loop bandwidth	f_{BW1}	High bandwidth option	—	1.6	—	MHz
	f_{BW2}	Low bandwidth option	—	475	—	kHz
MultiSynth frequency synthesis resolution	f_{RES}	Output frequency ≤ $F_{VCO}/8$	0	0	1	ppb
CLKIN loss of signal detect time	t_{LOS}		—	2.6	5	µs
CLKIN loss of signal release time	t_{LOSRLS}		0.01	0.2	1	µs
POR to output clock valid	t_{RDY}		—	—	2	ms
Input-to-output propagation delay	t_{PROP}	Buffer mode (PLL Bypass)	—	2.5	4	ns
Reset minimum pulse width	t_{RESET}		—	—	200	ns
Output-output skew[1]	t_{DSKEW}	F_{OUT} ≥ 5 MHz	—	—	100	ps
Spread spectrum PP frequency deviation[2]	SS_{DEV}	F_{OUT} = 100 MHz	—	–0.45	–0.5	%
Spread spectrum modulation rate[3]	SS_{DEV}	F_{OUT} = 100 MHz	30	31.5	33	kHz

1. Outputs at integer-related frequencies and using the same driver format.
2. Default value is 0.5% down spread.
3. Default value is 31.5 kHz for PCI compliance.

T_{PROP_ZDB} = Expected propagation delay in Zero Delay Buffer Mode = ±200 pS.

20. Can the Si5335 and Si5338 clock generators support spread spectrum inputs?

The Si5335 and Si5338 are generally used as “spread-aware” clock generators (i.e., PLLs that do not attenuate spread spectrum clocking at the input clocks). These products can be used as spread-aware clock generators as long as the input follows these conditions:

1. The input spread percentage should be within the tracking range of Si5335 and Si5338, which is 20,000 ppm. This translates to a maximum specification of ±2%.
2. The modulation rate cannot exceed 1.5 MHz.

If these two conditions are met, the Si5335 and Si5338 can be used as spread-aware clock generators.

21. Revision History

Revision	Date	Description
A	August, 2025	Initial Skyworks document release.

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